

**ATTACHMENT 4 – FEDERAL HIGHWAY ADMINISTRATION (FHWA)
TURNER-FAIRBANK HIGHWAY RESEARCH CENTER FINITE ELEMENT
DESIGN ANALYSIS REPORT**
(38 pages)

**FEDERAL HIGHWAY ADMINISTRATION
TURNER-FAIRBANK HIGHWAY RESEARCH CENTER REPORT**

I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation:

Finite Element Design Analysis

William Wright, Ph.D., P.E.
Federal Highway Administration

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PURPOSE

This model is created to perform detailed structural analysis of the tunnel roof panel system. The model shows how load is distributed throughout the system and provides an estimate of the load applied to each anchor. The model is also used to investigate the effect of anchor failures and to show how load redistribution occurs under different anchor-out scenarios. This information is intended to assist with engineering evaluation of the level of structural redundancy provided by the design.

Each component of the panel system is modeled using plate and beam elements to accurately recreate the geometry of the system. Model component weights are calibrated to match the measured component weights measured by the NTSB. The model is evaluated under both self weight (dead load) and with an imposed live load to help evaluate the factor of safety.

Redundancy of the system was investigated to determine the consequence of multiple anchor failures in the system. Pairs of anchors were systematically removed from the model to look at various “one out”, “two out”, and “three out” scenarios. Interpretation of the significance of the results is beyond the scope of this report.

MODEL DESCRIPTION

Figure 1 shows a rendering of the model with both the vertical and diagonal hanger rods installed. The model was created using FEMAP software and was executed using the static analysis module of ABAQUS version 6.6. Figure 2 isolates the EBD081-M1 hanger assembly that failed in the tunnel. The model was constructed according to plans provided by the NTSB⁽¹⁾. The following describes the key geometric features of the model:

- The WT hanger brackets connected to the tunnel roof are modeled using S4R shell elements as shown in figure 3. X-Y-Z translational constraints are placed on the nodes corresponding to the anchor locations.
- Support Beams M1 and M2 are modeled using S4R shell elements as shown in figure 4. The L shelf angles and ¾” x 12” hanger plate are modeled as separate components. Connections are provided between members at discrete locations corresponding to the bolt locations shown in the plan drawings. A single transverse shell element is used to simulate the effect of the bolted connections. The resulting hanger beams are therefore semi-composite, having greater stiffness than the three members separately, but having less stiffness than a rigidly connected member.

- WT Support Beams (N and S) are modeled using S4R shell elements as shown in figure 5. The side struts are rigidly connected to the WT beam flange to provide some torsional stiffness.
- The hanger rods are modeled as single beam elements with rotational releases at the ends where they are connected to the hanger plates. The actual hanger rods are assembled from multiple components (rod, clevis, turnbuckle, etc) with different stiffness. The composite stiffness of this assembly was measured in the laboratory in a load frame (See appendix 1). The hanger rods in the model are prismatic beam elements with a hollow tubular cross section. The size was chosen to match the axial stiffness measured in the laboratory.
- Concrete panels are modeled using 4" thick S4R shell elements. Figure 6 shows a cut-away view with one of the 12' x 8' panels installed. Ventilation holes are included in the panels where indicated. Each panel is discretely modeled and there is no connection between adjacent panels. The ends of each panel are connected to the tee beams and shelf angles at three points corresponding to the hold-down bolt locations. The connections are made using short beam elements to place the centroid of the panel at the correct location and to simulate a "semi-stiff" bolted connection.
- WT struts are provided in the model at the locations where the ceiling system was braced to the tunnel walls. The struts are pinned at the wall ends and rigidly fixed to the tee beam hangers. This provides some torsional bracing to the hanger beams to resist rotation under differential slab weight.
- The weight of the side plenum walls was simulated by placing a series of 50lb nodal loads on the concrete panels spaced every 2ft. along both sides of the ceiling system.

The model was run with two geometric variations of the hanger rods. There is some indication that the diagonal hanger rods were installed in the field after the concrete panels were placed in the system. Therefore, the level of load in the diagonal hanger rods due to the dead weight of the system will vary depending on how tight the rods were installed. This has some effect on the distribution of load on the individual anchors. Since the actual load is unknown, the model was run both with and without the diagonal hanger rods to provide an upper and lower bound for anchor loads. Figure 7 show the model with diagonal hanger rods removed.

For all steel material, $E = 29,700,000$ psi and $\mu = 0.32$ are the assumed elastic material properties. For concrete, the assumed material properties are $E=3,605,000$ psi and $\mu = 0.2$. Since this is linear elastic analysis, results are independent of material yield strength. Material density is discussed in the following section.

LOADS

Three load conditions were evaluated in the model as follows:

- Dead Load 1 (DL1): The model was analyzed under gravity with all diagonal hanger rods removed from the system. This assumes that the diagonal hanger rods are not installed until after the concrete panels are in place.
- Dead Load 2 (DL2): The model was analyzed under gravity with all diagonal hanger rods present in the system. Note that the summation of anchor bolt loads is slightly higher in this configuration due to the weight of the additional hanger rods.
- Live Load (LL): A uniform pressure of 25 psf. was applied in a downward vertical direction to the surface of the concrete panels. All diagonal hanger rods are present in the system for this load case.

Results are presented for each load case separately. Combined effects can be determined by superposition of the individual load case results. The combination DL1 + LL represents the construction sequence where the diagonal rods are installed with “zero” initial tension after the concrete panels are placed in the system. The DL2 + LL combination represents the construction sequence where all hanger rods are present and properly adjusted prior to placement of the concrete panels. Assuming proper hanger rod installation, these two combinations should form the upper and lower bounds for anchor load. For this report, all non-linear features were eliminated from the model to enable make superposition valid. This model does not consider the case on improper hanger rod installation where one or more hanger rods are misadjusted to carry disproportionate amounts of load.

The model component weights were calibrated to the actual component weights measured by the NTSB by making minor changes to the material density of steel and concrete. The final densities are $\gamma_{\text{steel}} = 536.3$ pcf. and $\gamma_{\text{concrete}} = 152.9$ pcf. This compares to the theoretical weights of 490 pcf. and 150 pcf., respectively, for these materials. This difference accounts for the weight of miscellaneous hardware and any dimensional tolerance differences for the actual components. Table 1 shows the model weight is within 0.9 % of the estimated weight based on component weights.

Table 1 Comparison of model weight to measured weights.

COMPONENT	UNIT WEIGHT (lbs)	QUANTITY	TOTAL (lbs)
12' x 8' Panel	4,865	4	19,460
12' x 8' Panel with Cutout	4,640	6	27,900
6' x 8' Panel	2,483	5	12,415
4 Bolt Hanger Assy.	163	6	978
2 Bolt Hanger Assy.	46	26	1,196
¾" x 12" Hanger Assy.	3,200	2	6,400
WT 13.5x42 Hanger Assy.	1,800	2	3,600
North WT Strut(1)	16.5	47.6 LF	785.4
South WT Strut(2)	25	95.9 LF	2,398
2' Ventilation Panel	50	40	2,000
SUM MEASURED COMPONENT WEIGHTS			77,532
SUM MODEL BOLT LOADS, (DL2)			76,817

(1) Not measured, assumed weight of WT 5 x 16.5

(2) Not measured, assumed Weight of WT 6 x 25

BASIC STRUCTURAL ANALYSIS

Figure 8 shows the deflected shape of the model with the diagonal hanger rods in place under self weight (DL2). The maximum vertical deflection is 0.17 in. occurring at about mid-span of the large concrete panels. Most of this deflection is due to bending of the panels, the deflection of the hanger plates is insignificant. The stress distribution in each element of the system was examined and found to be insignificant, also. Figure 8 also shows the vector sum of the reaction components present at each anchor location. Note that these values include the x, y, and z force components and they differ slightly from the tabulated load values that only show the vertical (z) component.

As indicated in figures 2 and 9, there were eight WT hanger brackets anchored to the roof to support hanger line EBD081-M1. These brackets were numbered from east to west according to the hanger rods that were attached to them. Brackets 1, 5, 6, 7, 8, and 12 have a single vertical hanger rod while brackets 2/3/4 and 9/10/11 have one vertical and two diagonal hanger rods. The brackets supporting the single vertical hanger rods were attached to the tunnel roof by two anchors at the north and south (N and S) ends of the brackets. The brackets supporting three hanger rods were attached to the tunnel roof by four anchors located at the corners of the hanger bracket (NE, NW, SE, and SW).

The axial load (z-axis load) in each anchor is tabulated in tables 2 through 5 for each hanger line in the system for the three basic load conditions (DL1, DL2, and LL). In addition, two load combinations (DL1 + LL, and DL2 + LL) are tabulated. For completeness, the strut end reactions are shown in figure 6 to enable force summation checks of the overall model. As previously mentioned, the anchor load distribution under

dead load depends on when the diagonal hanger rods are present and with how much tension. The DL1 case simulates the load distribution that would be present if the diagonal hanger rods are not present in the system when the concrete panels are placed. The DL2 case simulates the case where all hanger rods are present in the system and properly adjusted prior to placement of the concrete panels. Note that the summation of anchor loads in the DL1 case is 96 lbs less than the DL2 case due to the absence of the diagonal hanger rods.

The location of the maximum anchor load in hanger line M1 under dead load is dependent on the load case (DL1 versus DL2). With the diagonal hanger rods out (DL1), the maximum anchor load occurs at the west end of the hanger line. Under DL2, the maximum anchor load occurs at location 6 close to mid span. Based on field measurements, the hanger beams cantilever out 6 in. further on the west end compared to the east end. This creates a slight asymmetry to the anchor load distribution. In general, the highest anchor loads occur in the west half of the system. Hanger line EBD081-M1 has the highest anchor loads as expected.

REDUNDANCY ANALYSIS

The structural redundancy of the system was examined by removing a pattern of anchors from the system and repeating the analysis under both live load and dead load. Results were only evaluated for the DL2 and LL load cases with all diagonal hanger rods present in the system. Once anchors begin to fail, it can be assumed that load will redistribute among the hanger rods and any effect of hanger rod installation sequence becomes unimportant. Therefore, the DL1 load case was not considered for redundancy analysis. Hanger line EBD081-M1 has the highest proportion of load and is the hanger line that failed in service. Therefore, anchor and structural loads would be expected to be lower in other hanger lines, and failure scenarios involving anchors from these other lines were not considered.

A failure of a hanger rod anchorage was simulated by removing all of the anchor attachments from a given WT hanger bracket (either two or four anchors) and recalculating the load and stress distribution in the remaining structure. The removal of all anchors at a given hanger anchorage provides a “worst-case” for redundancy evaluation. Simulations involving removal of single hangers were not investigated.

This report focuses on three key results: 1) The redistribution of anchor loads; 2) the maximum Von Mises stress in the $\frac{3}{4}$ ” x 12” hanger plate generated by bending loads; and 3) the maximum downward deflection and its location. The model mesh is not designed to accurately evaluate local effects at connection points between the various components because there is no physical evidence that these local effects (such as bolted connection failure) played a significant role in the failure.

Stress results are reported in terms of the average Von Mises stress for the element where the maximum tensile membrane force occurs in the ¾" x 12" hanger plate. The average stress from both surfaces of the plate elements are reported to eliminate local "out-of-plane" bending effects from the results. The Von Mises stress provides the best measure for predicting failure where biaxial stress conditions exist. Maximum compressive bending stress was not considered critical since the hanger plate is well braced against local stability failure.

The anchor loads calculated for each scenario can be compared to the allowable epoxy anchor strength of 6,350 lbs. that was assumed in the design of the ceiling hanger system. ⁽²⁾ At the time the system was designed, the allowable anchor strength was calculated from the average ultimate anchor strength reported by the epoxy manufacturer using a safety factor of 4. The ultimate anchor strength would therefore have been 25,400 lbs. The steel properties of the hanger plate are not exactly known, but it can conservatively be assumed as ASTM grade A36 material (36,000 psi yield strength). It can also be assumed that the allowable design strength for A36 steel in bending will be 0.55 F_y or 19,800 psi. (this value may vary slightly depending on which code was used for design). This value can be compared to the Von Mises stress caused by bending of the hanger plate in the model.

"One-Out" Scenarios

Eight different "one-out" scenarios were investigated where all anchors at each of the eight hanger anchorage locations were removed sequentially. Each case, therefore, had either two or four anchors removed from the system. Results, shown in table 7, show the maximum anchor load and location for each scenario. In addition, the maximum Von Mises stress in the hanger plate is shown along with its location (distance from the west end of the ceiling system). The maximum deflection is also reported for each scenario.

The maximum anchor load occurred at location 12 when the anchors at location 9/10/11 were removed from the system. The maximum anchor load (DL2 + LL: 4,412 lbs) is below the allowable design load for the anchors (6,350 lbs). Therefore, it can be concluded that the allowable design capacity of the anchors will not be exceeded in any of the "one-out" scenarios.

The maximum bending stress in the hanger plate occurs 45 in. from the west end of the ceiling system when the anchors were removed from location 12. Since the calculated stress (DL2 + LL: 3,495 psi.) is well below the allowable design stress (19,800 psi), yielding of the hanger plate will not occur in any of the "one-out" scenarios.

The maximum vertical deflection of the hanger plate under dead load occurs at the west end when the anchors are removed from location 12. The deflection only increased by 0.051 in. compared to the dead load deflection of the undamaged structure. This indicates there would be no appreciable visual warning from below the ceiling that a "one-out" condition was present in the system.

In summary, the ceiling system is expected to maintain its full design safety capacity under any “one-out” anchor failure scenario.

“Two-Out” Scenarios

Seven different “two-out” scenarios were investigated where all anchors at adjacent pairs of hanger anchorage locations were removed sequentially. Each case, therefore, had either four or six anchors removed from the system. Results, shown in table 8, show the maximum anchor load and location for each scenario. In addition, the maximum Von Mises stress in the hanger plate is shown along with its location (distance from the west end of the ceiling system). The maximum deflection is also reported for each scenario.

The maximum anchor load occurred at location 8 when the anchors at locations 9/10/11 and 12 were removed from the system. The maximum anchor load (DL2 + LL: 10,887 lbs) exceeds the allowable design load for the anchors (6,350 lbs), but does not exceed the ultimate capacity of the anchors. Therefore, it can be concluded that the structure has a reduced factor of safety (2.3 versus 4) with respect to anchor failure in the worst case “two-out” scenario.

The maximum bending stress in the hanger plate occurs 159 in. from the west end of the ceiling system when the anchors were removed from locations 9/10/11 and 12. Since the calculated stress (DL2 + LL: 17,513 psi.) is below the allowable design stress (19,800 psi), yielding of the hanger plate will not occur in any of the “two-out” scenarios.

The maximum vertical deflection of the hanger plate under dead load occurs at the west end when the anchors are removed from locations 9/10/11 and 12. The deflection increased by 0.987 in. compared to the dead load deflection of the undamaged structure. Depending on conditions, this deflection may have been visually detectable from below the ceiling system.

In summary, under any “two-out” scenario, the ceiling system is expected to maintain its full design factor of safety with respect to yielding but there is a 43% reduction in the factor of safety against anchor failure.

“Three-Out” Scenarios

Six different “three-out” scenarios were investigated where all anchors at adjacent triplets of hanger anchorage locations were removed sequentially. Each case, therefore, had either six or eight anchors removed from the system. Results, shown in table 9, show the maximum anchor load and location for each scenario. In addition, the maximum Von Mises stress in the hanger plate is shown along with its location (distance from the west end of the ceiling system). The maximum deflection is also reported for each scenario.

The maximum anchor load occurred at location 7 when the anchors at locations 8, 9/10/11, and 12 were removed from the system. The maximum anchor load (DL2 + LL: 14,811 lbs) exceeds the allowable design load for the anchors (6,350 lbs), but does not exceed the ultimate capacity of the anchors. Therefore, it can be concluded that the structure has a reduced factor of safety (1.7 versus 4) with respect to anchor failure in the worst case “three-out” scenario.

The maximum bending stress in the hanger plate occurs 225 in. from the west end of the ceiling system when the anchors were removed from locations 8, 9/10/11, and 12. Since the calculated stress (DL2 + LL: 26,066 psi.) exceeds the allowable design stress (19,800 psi) but does not exceed the yield strength (36,000 psi), it can be concluded that yielding will not occur but the design factor of safety will be reduced by 31% under the worst-case “three-out” scenario. It should be noted that the calculated stress does not exceed the allowable design stress for ASTM grade 50 steel (27,500 psi), which is a common material for this application. Therefore, it may be possible that the actual hanger plate meets the full factor of safety with respect to bending, but the material must be verified before this conclusion can be made.

The maximum vertical deflection of the hanger plate under dead load occurs at the west end when the anchors are removed from locations 8, 9/10/11 and 12. The deflection increased by 2.132 in. compared to the dead load deflection of the undamaged structure. Depending on conditions, this deflection may have been visually detectable from below the ceiling system.

In summary, under any “three-out” scenario, the ceiling system has a 31% reduction in the factor of safety with respect to bending and a 58% reduction in the factor of safety with respect to anchor failure.

Redundancy Summary

The following conclusions are based on the assumed allowable design strengths of 6,350 lbs. for the anchor capacity and 19,800 psi. for the bending capacity of the hanger plate. Changing these values will alter the conclusions.

The analysis shows that the ceiling system maintains its full factor of safety with respect to anchor failure and bending failure under the worst-case “one-out” scenario. Therefore, the system can be considered to have full “one-out” redundancy.

The analysis shows that the ceiling system has a 43% reduction in the factor of safety with respect to anchor failure under the worst-case “two-out” scenario, but it maintains its full factor of safety with respect to bending failure. Therefore, the system cannot be considered to have full “two-out” redundancy. However, the system still has significant capacity under “two-out” conditions and collapse would not be expected.

Under the worst “three-out” scenario, the system cannot be considered redundant since both the allowable anchor capacity and allowable bending capacity of the hanger plate are exceeded. The system may have sufficient capacity to avoid collapse if the anchors are capable of achieving their rated ultimate capacity. However, a more detailed analysis of local hot-spot stress and bolted connection capacity needs to be performed before any conclusions can be made about collapse.

REFERENCES

- 1) Record Drawings, Massachusetts Highway Department - Central Artery (I-93)/Tunnel (I-90) Project – I-90 Seaport Access Tunnel Finishes – I-90 E. BD. Ceiling Module Group Plan. January 9, 2004.
- 2) NTSB Structures Group Chairman’s Factual Report.

Table 2 Bolt Load for Hanger Line EBD081-N

LOCATION	BOLT ID	BOLT LOAD (DL1) DIAGONALS OUT (lbs)	BOLT LOAD (DL2) DIAGONALS IN (lbs)	BOLT LOAD (LL) DIAGONALS IN (lbs)	BOLT LOAD (DL1 + LL) (lbs)	BOLT LOAD (DL2 + LL) (lbs)
1	N	851	859	328	1,179	1,187
	S	851	859	328	1,179	1,187
2	N	986	882	334	1,321	1,216
	S	986	882	334	1,321	1,216
3	N	980	772	291	1,271	1,063
	S	980	772	291	1,271	1,063
4-5-6	NE	569	888	332	901	1,220
	SE	569	888	332	901	1,220
	NW	569	826	307	875	1,132
	SW	569	826	307	875	1,132
7	N	902	730	276	1,178	1,005
	S	902	730	276	1,178	1,005
8	N	940	852	324	1,264	1,176
	S	940	852	324	1,264	1,176
9	N	973	966	368	1,341	1,334
	S	973	966	368	1,341	1,334
10	N	1,248	1,251	482	1,730	1,733
	S	1,248	1,251	482	1,730	1,733
SUM		16,035	16,050	6,082	22,117	22,132

Table 3 Bolt Load for Hanger Line EBD081-M1

LOCATION	BOLT ID	BOLT LOAD DL(1) DIAGONALS OUT (lbs)	BOLT LOAD DL(2) DIAGONALS IN (lbs)	BOLT LOAD LL DIAGONALS IN (lbs)	BOLT LOAD DL(1) + LL (lbs)	BOLT LOAD DL(2) + LL (lbs)
1	N	1,589	1,086	466	2,055	1,552
	S	1,589	1,086	466	2,055	1,552
2-3-4	NE	911	1,387	584	1,494	1,971
	SE	911	1,387	584	1,494	1,971
	NW	911	1,375	572	1,483	1,947
	SW	911	1,375	572	1,483	1,947
5	N	1,586	1,280	539	2,125	1,819
	S	1,586	1,280	539	2,125	1,819
6	N	1,795	1,665	706	2,501	2,371
	S	1,795	1,665	706	2,501	2,371
7	N	1,597	1,471	620	2,217	2,091
	S	1,597	1,471	620	2,217	2,091
8	N	1,525	1,199	504	2,029	1,703
	S	1,525	1,199	504	2,029	1,703
9-10-11	NE	974	1,474	616	1,590	2,090
	SE	974	1,474	616	1,590	2,090
	NW	974	1,553	655	1,630	2,208
	SW	974	1,553	655	1,630	2,208
12	N	2,159	1,549	664	2,823	2,213
	S	2,159	1,549	664	2,823	2,213
SUM		28,044	28,078	11,853	39,897	39,931

Table 4 Bolt Load for Hanger Line EBD081-M2

LOCATION	BOLT ID	BOLT LOAD (DL1) DIAGONALS OUT (lbs)	BOLT LOAD (DL2) DIAGONALS IN (lbs)	BOLT LOAD (LL) DIAGONALS IN (lbs)	BOLT LOAD (DL1 + LL) (lbs)	BOLT LOAD (DL2 + LL) (lbs)
1	N	1,265	864	353	1,618	1,217
	S	1,265	864	353	1,618	1,217
2-3-4	NE	730	1,110	442	1,172	1,552
	SE	730	1,110	442	1,172	1,552
	NW	730	1,105	435	1,165	1,540
	SW	730	1,105	435	1,165	1,540
5	N	1,272	1,026	410	1,682	1,436
	S	1,272	1,026	410	1,682	1,436
6	N	1,449	1,344	543	1,992	1,887
	S	1,449	1,344	543	1,992	1,887
7	N	1,262	1,161	462	1,724	1,623
	S	1,262	1,161	462	1,724	1,623
8	N	1,219	960	383	1,602	1,343
	S	1,219	960	383	1,602	1,343
9-10-11	NE	778	1,175	465	1,243	1,640
	SE	778	1,175	465	1,243	1,640
	NW	778	1,235	494	1,272	1,729
	SW	778	1,235	494	1,272	1,729
12	N	1,706	1,225	499	2,205	1,724
	S	1,706	1,225	499	2,205	1,724
SUM		22,379	22,411	8,970	31,349	30,362

Table 5 Bolt Load for Hanger Line EBD081-S

LOCATION	BOLT ID	BOLT LOAD DL(1) DIAGONALS OUT (lbs)	BOLT LOAD DL(2) DIAGONALS IN (lbs)	BOLT LOAD LL DIAGONALS IN (lbs)	BOLT LOAD DL(1) + LL (lbs)	BOLT LOAD DL(2) + LL (lbs)
1	N	523	524	150	673	774
	S	523	524	150	673	774
2	N	640	571	166	806	737
	S	640	571	166	806	737
3	N	644	504	144	788	648
	S	644	504	144	788	648
4-5-6	NE	378	604	176	554	780
	SE	378	604	176	554	780
	NW	378	521	142	521	663
	SW	378	521	142	521	663
7	N	575	475	136	711	612
	S	575	475	136	711	612
8	N	592	544	155	747	699
	S	592	544	155	747	699
9	N	613	611	173	786	783
	S	613	611	173	786	783
10	N	787	785	226	1,013	1,011
	S	787	785	226	1,013	1,011
SUM		10,263	10,278	2,857	13,199	13,214

Table 6 Reaction Load for Strut Ends

LOCATION	STRUT ID	BOLT LOAD DL(1) DIAGONALS OUT (lbs)	BOLT LOAD DL(2) DIAGONALS IN (lbs)	BOLT LOAD LL DIAGONALS IN (lbs)	BOLT LOAD DL(1) + LL (lbs)	BOLT LOAD DL(2) + LL (lbs)
SOUTH WALL	1	226	226	-8	218	218
	2	233	232	-8	225	225
	3	240	240	-7	232	232
	4	247	247	-7	240	240
	5	256	256	-7	249	249
SUM		1,201	1,202	-36	1,165	1,165
NORTH WALL	1	10	10	-35	-25	-24
	2	6	5	-37	-31	-32
	3	4	4	-37	-32	-32
	4	2	2	-37	-35	-35
	5	2	2	-37	-35	-35
SUM		24	24	-182	-158	-158

Table 7 “One Out” Scenarios on Hanger Line EBD081-M1

BOLTS REMOVED	MAXIMUM BOLT LOAD			MAXIMUM DEFLECTION			MAXIMUM PLATE BENDING STRESS (Von Mises)		
	DL2 (lbs)	LL (lbs)	BOLT LOCATIONS	DL2 (in)	LL (in)	DISTANCE FROM WEST END (in)	DL2 (psi)	LL (psi)	DISTANCE FROM WEST END (in)
None	1,665	706	6-N 6-S	-0.042	-0.018	6	2,173	989	45
1-N 1-S	2,355	999	2/3/4-NE 2/3/4-SE	-0.046	-0.020	474	2,199	999	45
2/3/4-NE 2/3/4-SE 2/3/4-NW 2/3/4-SW	2,473	1,050	1-N 1-S	-0.048	-0.020	396	2,172	989	45
5-N 5-S	2,099	889	6-N 6-S	-0.043	-0.019	6	2,159	985	45
6-N 6-S	2,149	907	7-N 7-S	-0.050	-0.021	282	2,158	983	45
7-N 7-S	2,295	971	6-N 6-S	-0.046	-0.019	216	2,217	967	159
8-N 8-S	2,090	875	9/10/11-NE 9/10/11-SE	-0.038	-0.017	6	2,206	1,003	45
9/10/11-NE 9/10/11-SE 9/10/11-NW 9/10/11-SW	3,095	1,317	12-N 12-S	-0.057	-0.025	6	2,188	992	45
12-N 12-S	2,929	1,246	9/10/11-NW 9/10/11-SW	-0.093	-0.040	6	2,411	1,084	45

Table 8 “Two Out” Scenarios on Hanger Line EBD081-M1

BOLTS REMOVED	MAXIMUM BOLT LOAD			MAXIMUM DEFLECTION			MAXIMUM PLATE BENDING STRESS (Von Mises)		
	DL2 (lbs)	LL (lbs)	BOLT LOCATIONS	DL2 (in)	LL (in)	DISTANCE FROM WEST END (in)	DL2 (psi)	LL (psi)	DISTANCE FROM WEST END (in)
None	1,665	706	6-N 6-S	-0.042	-0.018	6	2,173	989	45
1-N 1-S 2/3/4-NE 2/3/4-SE 2/3/4-NW 2/3/4-SW	6,698	2,830	5-N 5-S	-0.770	-0.327	474	11,152	4,558	327
2/3/4-NE 2/3/4-SE 2/3/4-NW 2/3/4-SW 5-N 5-S	3,830	1,612	6-N 6-S	-0.108	-0.045	360	2,204	991	45
5-N 5-S 6-N 6-S	3,010	1,263	2/3/4-NW 2/3/4-SW	-0.073	-0.031	294	2,130	974	45
6-N 6-S 7-N 7-S	2,414	1,019	5-N 5-S	-0.092	-0.039	252	3,399	1,456	159
7-N 7-S 8-N 8-S	2,938	1,232	9/10/11-NE 9/10/11-SE	-0.067	-0.028	204	2,164	987	45
8-N 8-S 9/10/11-NE 9/10/11-SE 9/10/11-NW 9/10/11-SW	3,762	1,596	12-N 12-S	-0.111	-0.047	138	2,867	1,212	219
9/10/11-NE 9/10/11-SE 9/10/11-NW 9/10/11-SW 12-N 12-S	7,648	3,239	8-N 8-S	-1.029	-0.438	6	12,332	5,181	159

Table 9 “Three Out” Scenarios on Hanger Line EBD081-M1

BOLTS REMOVED	MAXIMUM BOLT LOAD			MAXIMUM DEFLECTION			MAXIMUM PLATE BENDING STRESS (Von Mises)		
	DL2 (lbs)	LL (lbs)	BOLT LOCATIONS	DL2 (in)	LL (in)	DISTANCE FROM WEST END (in)	DL2 (psi)	LL (psi)	DISTANCE FROM WEST END (in)
None	1,665	706	6-N 6-S	-0.042	-0.018	6	2,173	989	45
1-N 1-S 2/3/4-NE 2/3/4-SE 2/3/4-NW 2/3/4-SW 5-N 5-S	9,752	4,119	6-N 6-S	-1.724	-0.730	474	12,915	5,388	279
2/3/4-NE 2/3/4-SE 2/3/4-NW 2/3/4-SW 5-N 5-S 6-N 6-S	4,780	2,012	7-N 7-S	-0.265	-0.111	336	4,509	1,890	219
5-N 5-S 6-N 6-S 7-N 7-S	3,791	1,593	2/3/4-NW 2/3/4-SW	-0.122	-0.052	264	4,231	1,799	159
6-N 6-S 7-N 7-S 8-N 8-S	3,768	1,583	9/10/11-NE 9/10/11-SE	-0.120	-0.050	234	4,127	1,742	333
7-N 7-S 8-N 8-S 9/10/11-NE 9/10/11-SE 9/10/11-NW 9/10/11-SW	4,914	2,070	6-N 6-S	-0.250	-0.105	162	3,229	1,368	333
8-N 8-S 9/10/11-NE 9/10/11-SE 9/10/11-NW 9/10/11-SW 12-N 12-S	10,504	4,442	7-N 7-S	-2.174	-0.923	6	18,687	7,379	225

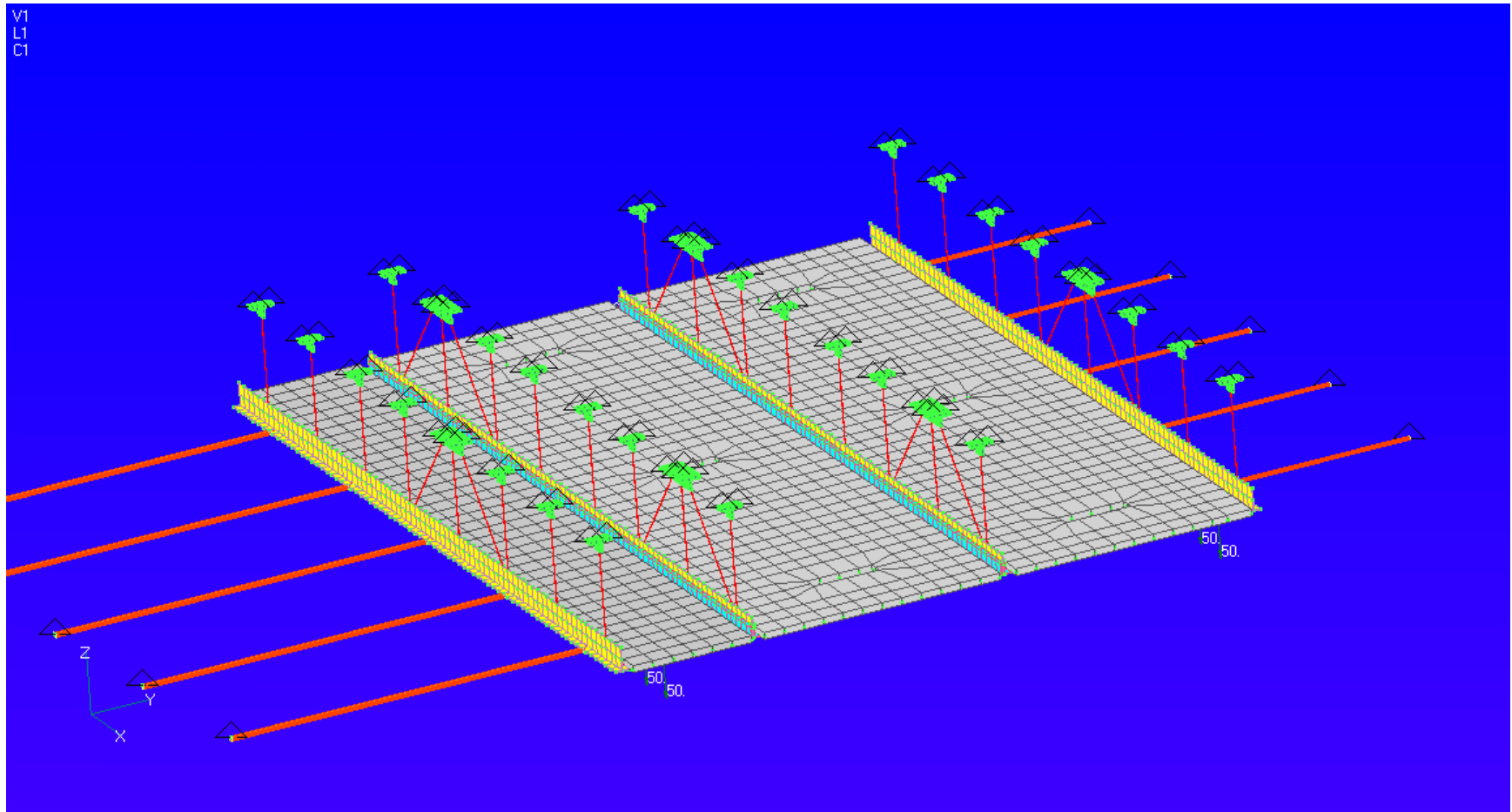


Figure 1 Finite Element Model of Panel Section EBD081

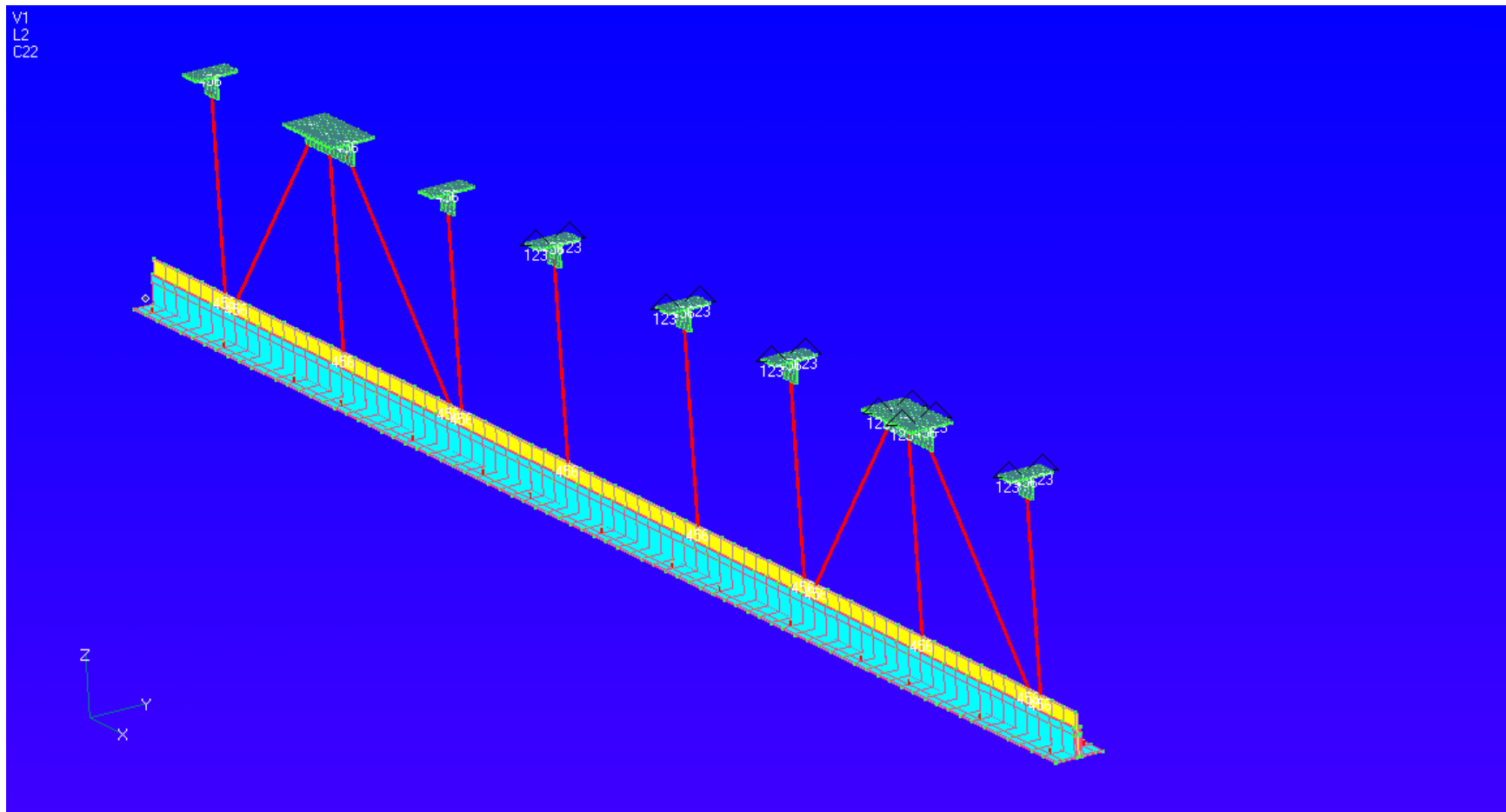


Figure 2 Hanger line EBD081-M1, showing the built-up beam assembly, hanger rods, and WT hanger brackets. Constraints are placed at all anchor locations.

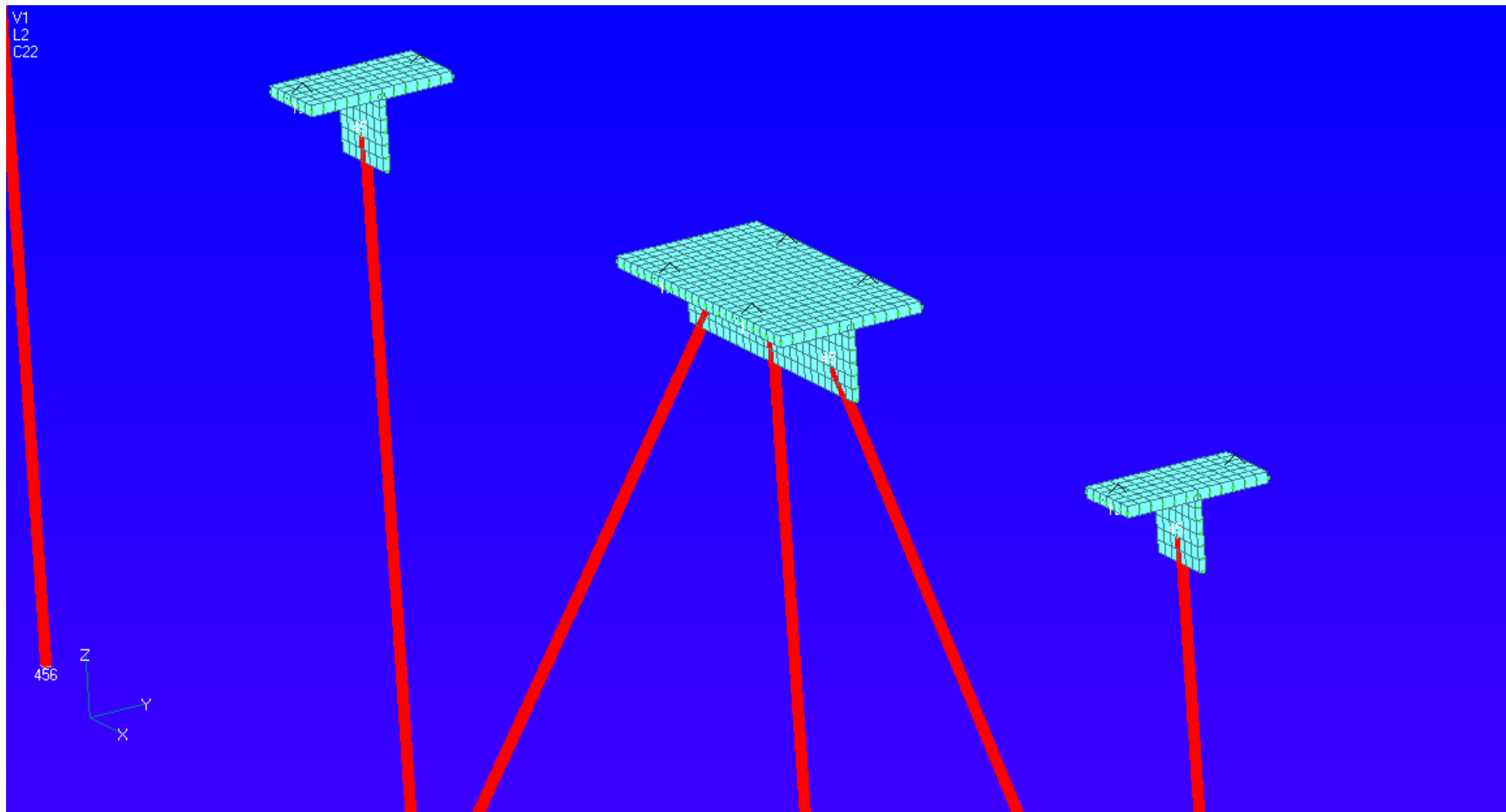


Figure 3 Close-up of the WT hanger brackets showing the constraints at the bolt locations.

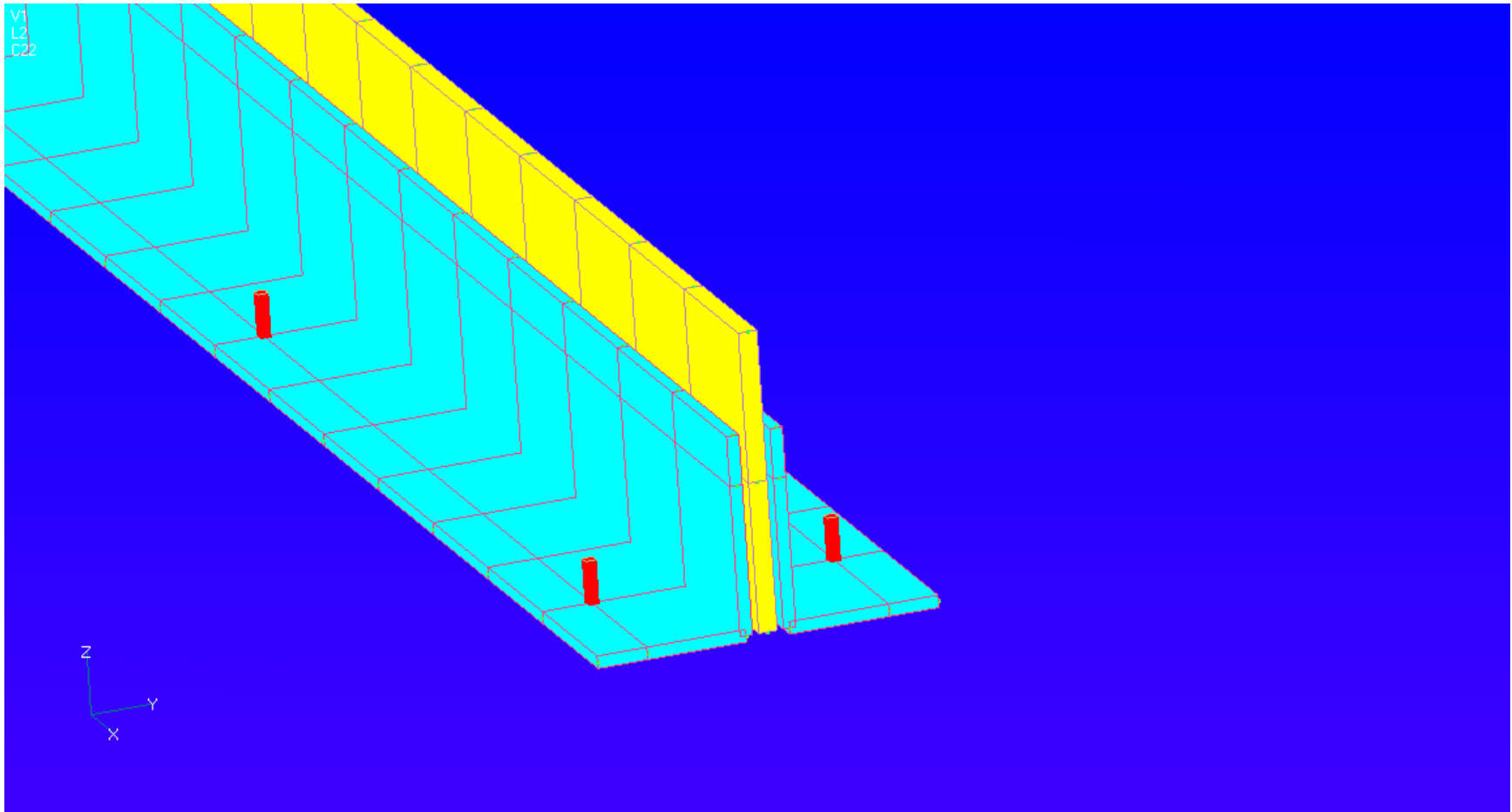


Figure 4 Close-up of the M1 and M2 built-up beam assemblies. The hanger plate and shelf angles are modeled as separate components and are connected together at discrete bolt locations. The studs used to connect the concrete panels are shown in red.

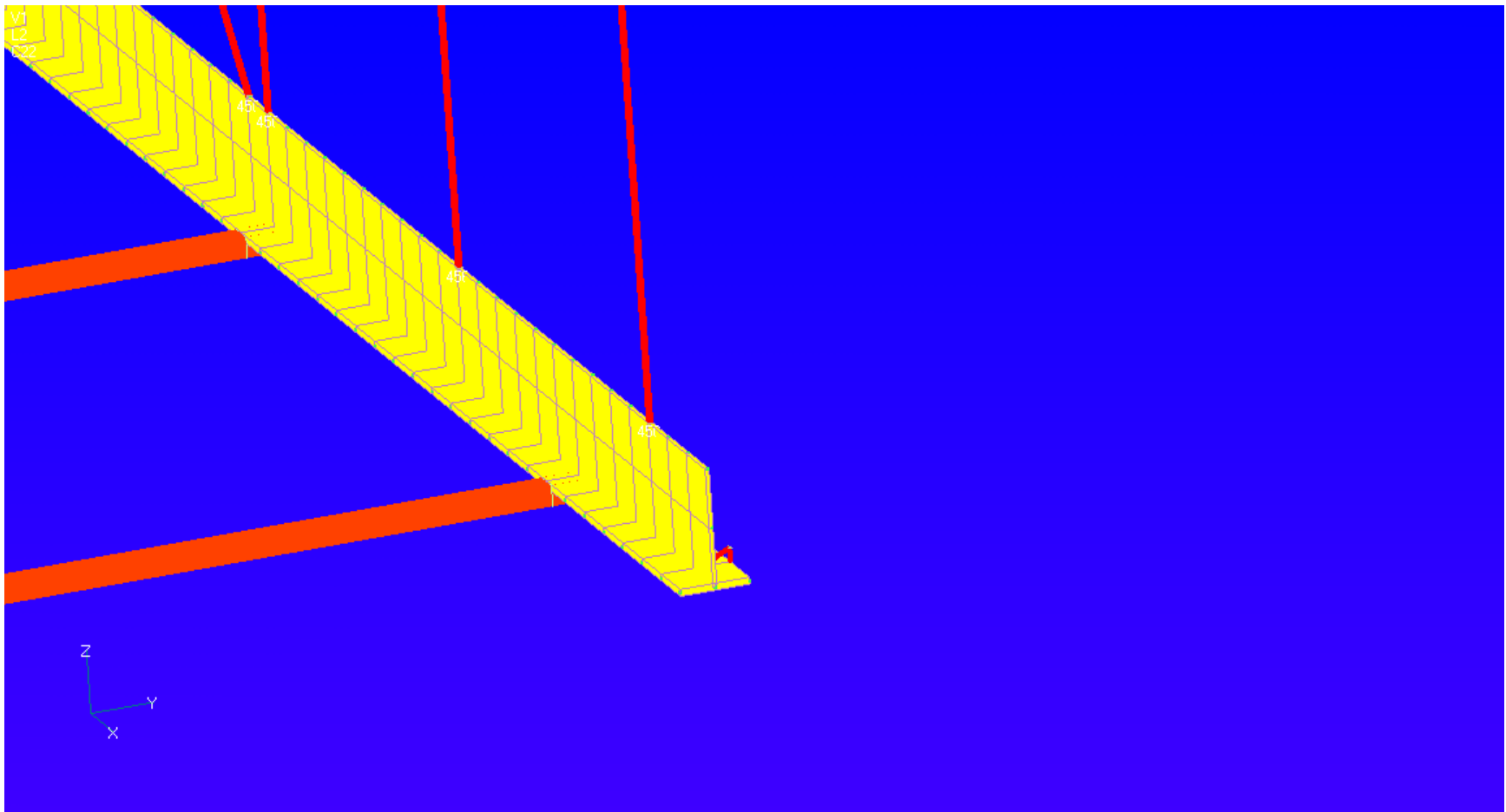


Figure 5 Hanger line EBD081-S showing the plate elements used for the WT beam and the connection of the struts bracing the assembly to the tunnel wall.

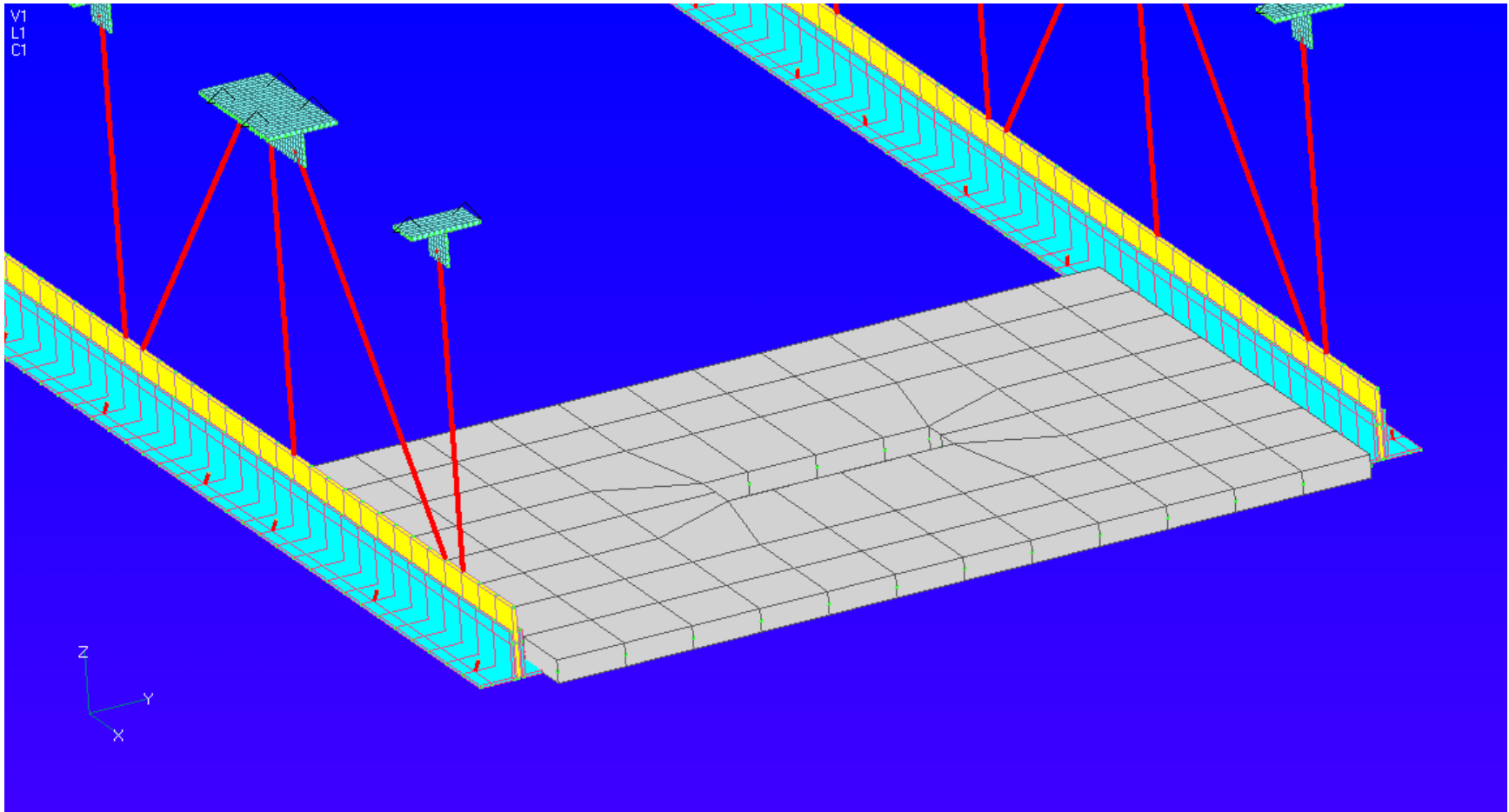


Figure 6 Cut-away showing one of the south concrete panels and its connection to hanger lines M1 and M2.

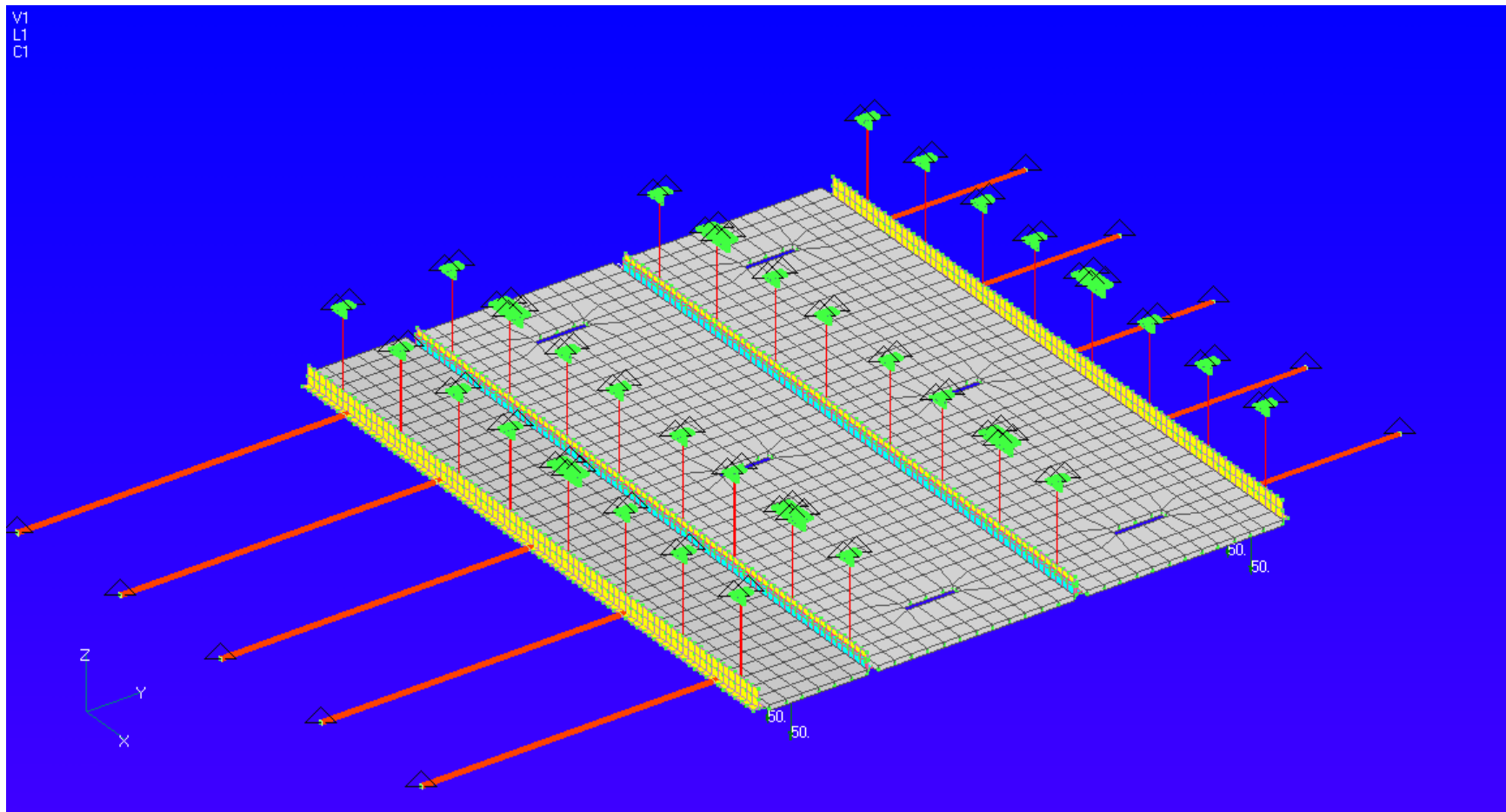


Figure 7 Model with all diagonal hanger rods removed for the DL1 load case.

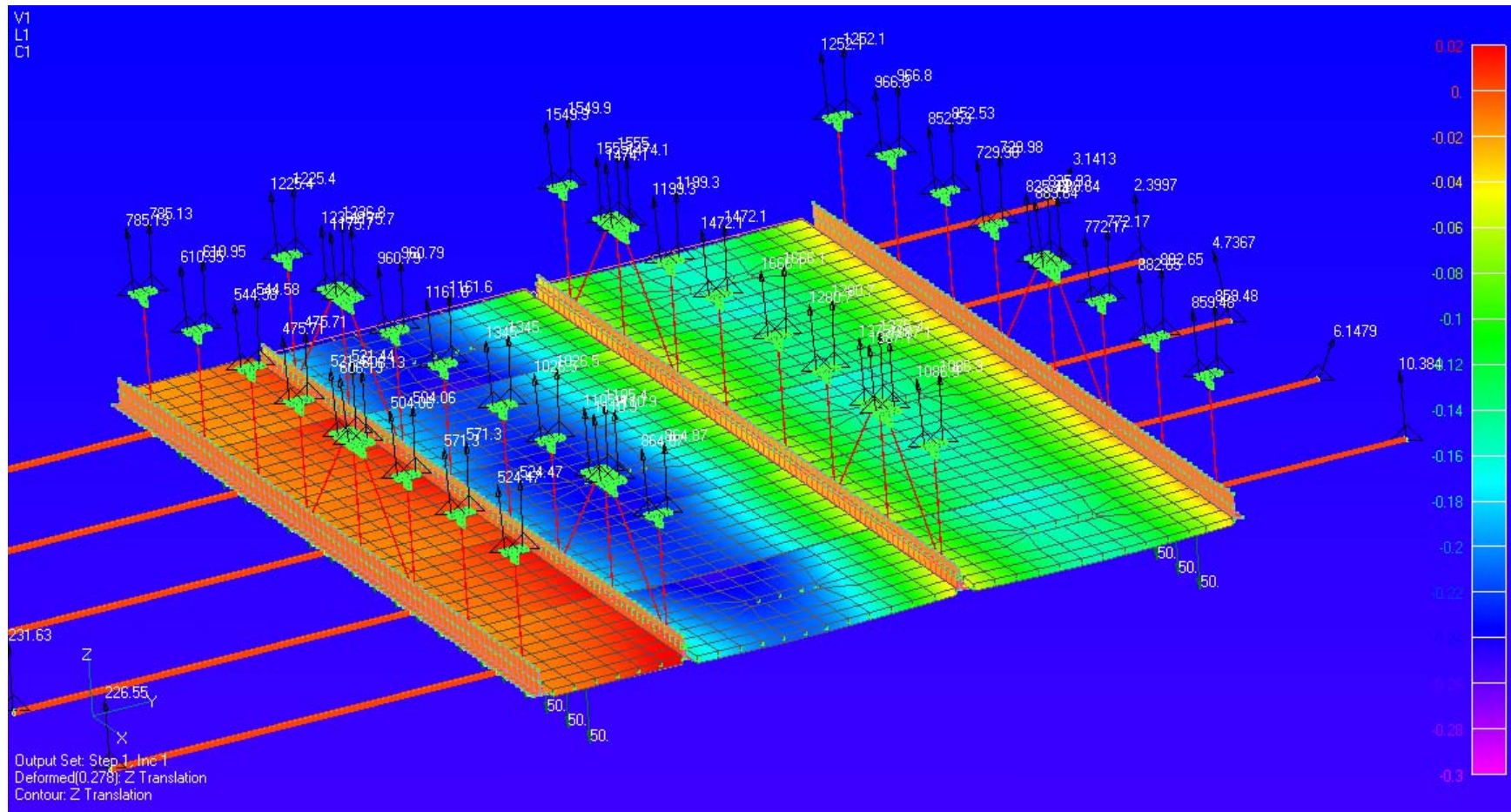


Figure 8 Deflected shape of the model under self weight with all bolts and hanger rods in place (DL2).

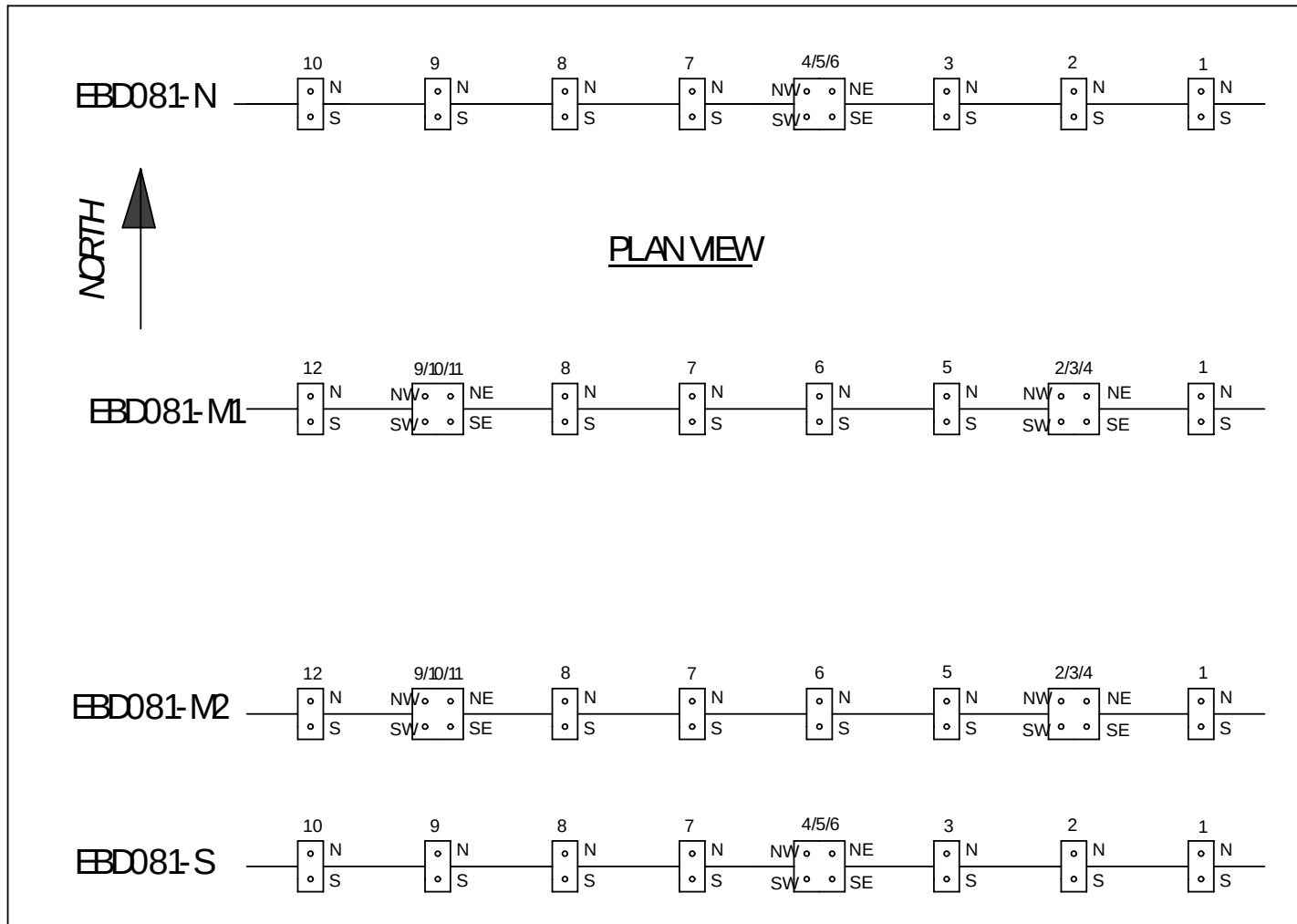


Figure 9 Plan view of the roof system showing the anchor labeling scheme.

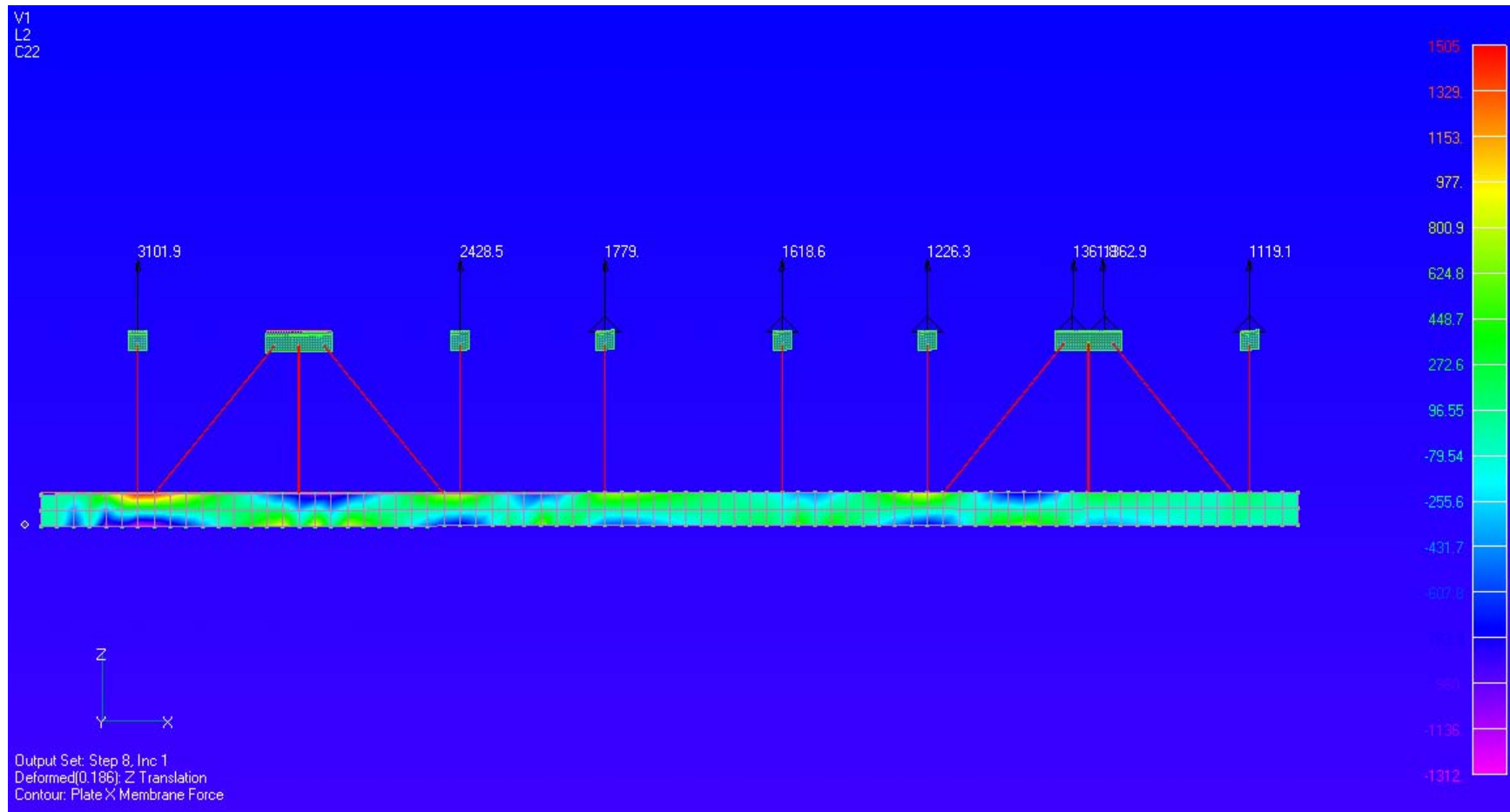


Figure 10 Worst-case "one-out" scenario showing the membrane force distribution in the 3/4" x 12" hanger plate under self weight (DL2).

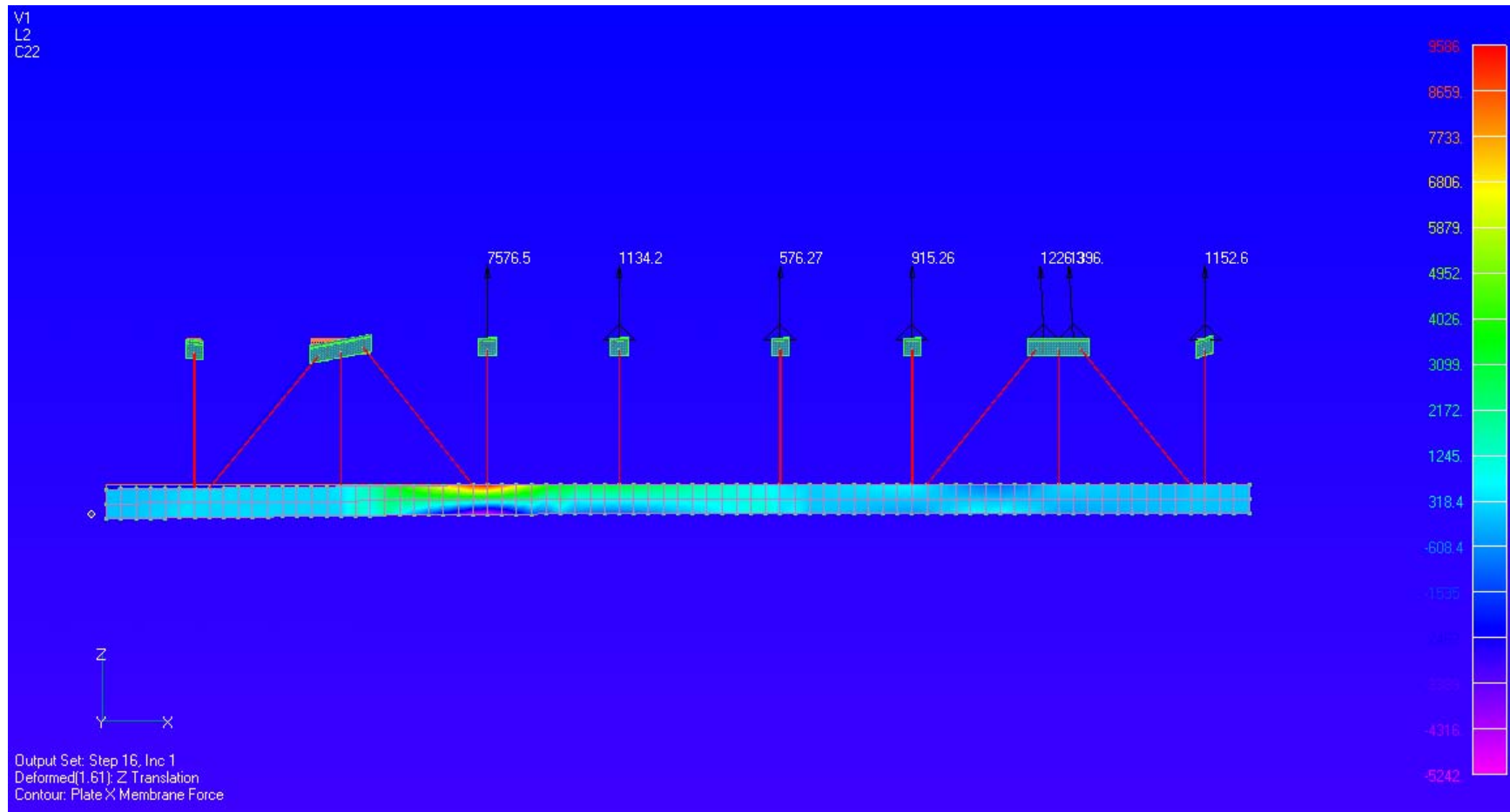


Figure 11 Worst-case "two out" scenario showing the maximum membrane force in the 3/4" x 12" hanger plate under self-weight (DL2).

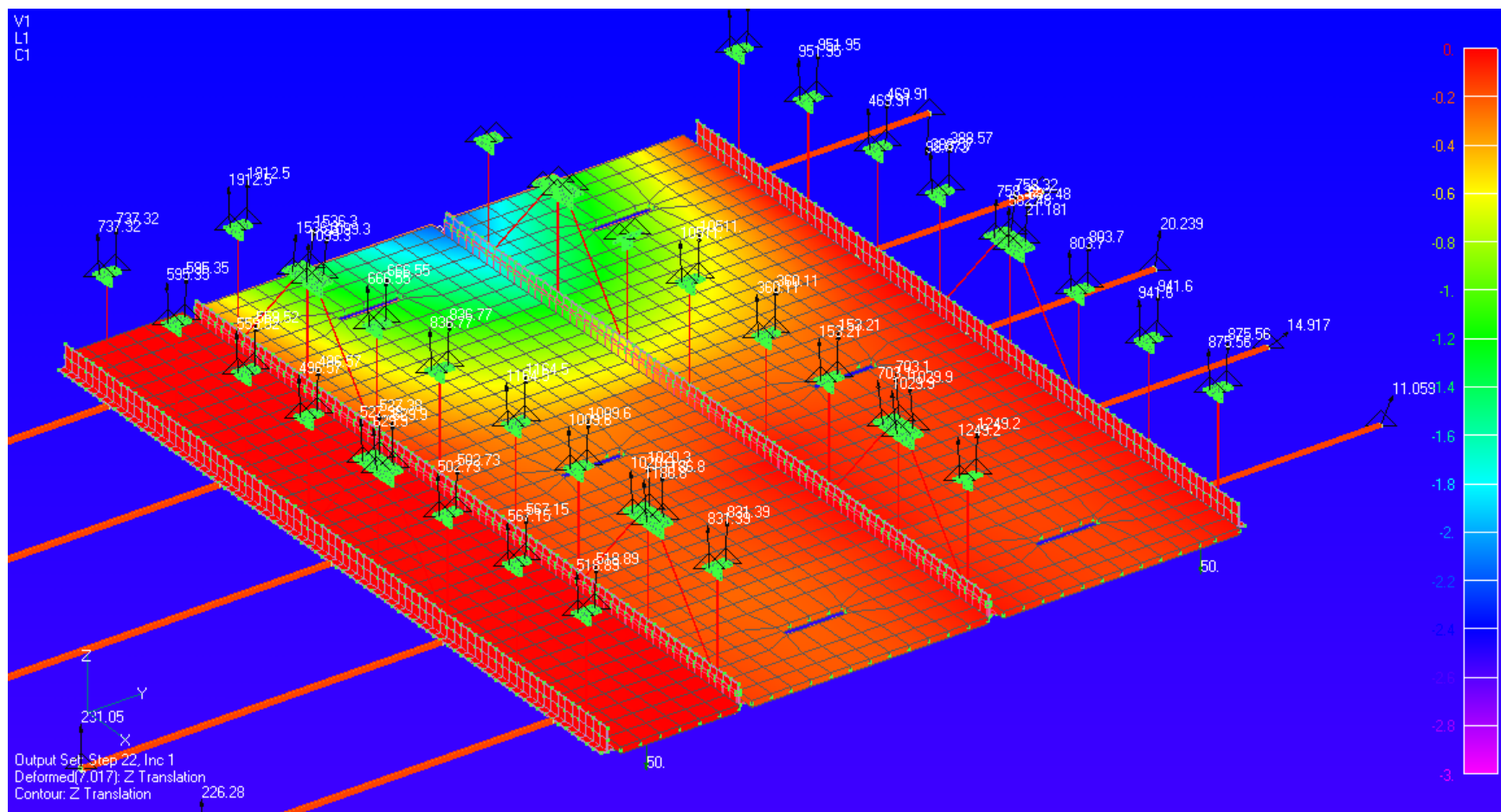


Figure 13 3D view of the worst-case "three out" scenario showing -2.17 in. dead load deflection at the west end (left) of hanger line M1.

APPENDIX 1 HANGER ROD STIFFNESS TESTS

Tests were performed on two undamaged hanger rods to determine the axial stiffness of the built-up assembly. The rods were tested in a 100kip MTS servo-hydraulic load frame using a pin and clevis grip system. The applied load is measured with a load cell and LVDTs were used to measure elongation over the entire gauge length. Figures A1 and A2 show the resulting stress versus strain behavior of the hanger rod assemblies. Stress is calculated by dividing load by the cross sectional area of the round bar portion of the assembly. Strain is determined by dividing the overall elongation of the assembly by the overall gauge length. Linear regression was used to determine the slope of the stress versus curve.

Prismatic beam elements with a 0.98in. diameter tubular cross section are used to model the hanger rods in the finite element model. The cross sectional area $A = 0.403 \text{ in}^2$. This results in a beam element with equivalent axial stiffness compared to the measured test results.

Summary of Tension Test Results

Data File: D:\Boston Tunnel\HCV-050-M-2a.csv

Steel design: ASTM A572

No.	Description of Results	SI Units		US Units	
		Value	Units	Value	Units
	<i>General:</i>				
1	Testing temperature	24.00	°C	75.20	°F
2	Specimen initial length	1625.60	mm	64.00	in
3	Specimen cross sectional area	433.55	mm ²	0.67	in ²
4	Measure rupture length	...	mm	...	in
5	Measure rupture cross sectional area	...	mm ²	...	in ²
	<i>Stresses:</i>				
6	Offset(0.2%) yield stress	...	MPa	...	Ksi
7	Ext. Under Load(0.5%) yield stress	...	MPa	...	Ksi
8	Average yield stress	...	MPa	...	Ksi
9	Upper yield stress	...	MPa	...	Ksi
10	Lower yield stress	...	MPa	...	Ksi
11	Ultimate stress	...	MPa	...	Ksi
12	Final stress	152.62	MPa	22.14	Ksi
13	Average static yield stress	...	MPa	...	Ksi
	<i>Strains:</i>				
14	Offset(0.2%) yield strain	...	%	...	%
15	Ext. Under Load(0.5%) yield strain	...	%	...	%
16	Average yield strain	...	%	...	%
17	Upper yield strain	...	%	...	%
18	Lower yield strain	...	%	...	%
19	Ultimate strain	...	%	...	%
20	Final strain	0.14	%	0.14	%
21	Average static yield strain	...	%	...	%
	<i>Young Modulus:</i>				
22	Least square method	123.15	GPa	17862.08	Ksi
23	Chord method	...	GPa	...	Ksi
24	Tangent method	...	GPa	...	Ksi
25	Correlation of LSQ method	0.99950	-	0.99950	-
	<i>Strain Hardening Modulus:</i>	...			
26	Apparent onset of hardening	...	%	...	%
27	Strain at hardening	...	%	...	%
28	Stress at hardening	...	MPa	...	Ksi

Table A1 Calculated data from hanger rod assembly test A.

Summary of Tension Test Results

Data File: D:\Boston Tunnel\HOV-080-N-2b.tfd

Steel desig.: ASTM A572

No.	Description of Results	SI Units		US Units	
		Value	Units	Value	Units
General:					
1	Testing temperature	24.00	°C	75.20	°F
2	Specimen initial length	1720.85	mm	67.75	in
3	Specimen cross sectional area	433.55	mm ²	0.67	in ²
4	Measure rupture length	...	mm	...	in
5	Measure rupture cross sectional area	...	mm ²	...	in ²
Stresses:					
6	Offset(0.2%) yield stress	...	MPa	...	Ksi
7	Ext. Under Load(0.5%) yield stress	...	MPa	...	Ksi
8	Average yield stress	...	MPa	...	Ksi
9	Upper yield stress	...	MPa	...	Ksi
10	Lower yield stress	...	MPa	...	Ksi
11	Ultimate stress	...	MPa	...	Ksi
12	Final stress	152.80	MPa	22.16	Ksi
13	Average static yield stress	...	MPa	...	Ksi
Strains:					
14	Offset(0.2%) yield strain	...	%	...	%
15	Ext. Under Load(0.5%) yield strain	...	%	...	%
16	Average yield strain	...	%	...	%
17	Upper yield strain	...	%	...	%
18	Lower yield strain	...	%	...	%
19	Ultimate strain	...	%	...	%
20	Final strain	0.12	%	0.12	%
21	Average static yield strain	...	%	...	%
Young Modulus:					
22	Least square method	123.37	GPa	17893.36	Ksi
23	Chord method	...	GPa	...	Ksi
24	Tangent method	...	GPa	...	Ksi
25	Correlation of LSQ method	0.99966	-	0.99966	-
Strain Hardening Modulus:					
26	Apparent onset of hardening	...	%	...	%
27	Strain at hardening	...	%	...	%
28	Stress at hardening	...	MPa	...	Ksi
29	Modulus hardening	...	GPa	...	Ksi

Figure Calculated data from hanger rod assembly test B.

Stress-Strain Curve of Hanger HOV-080-N2a

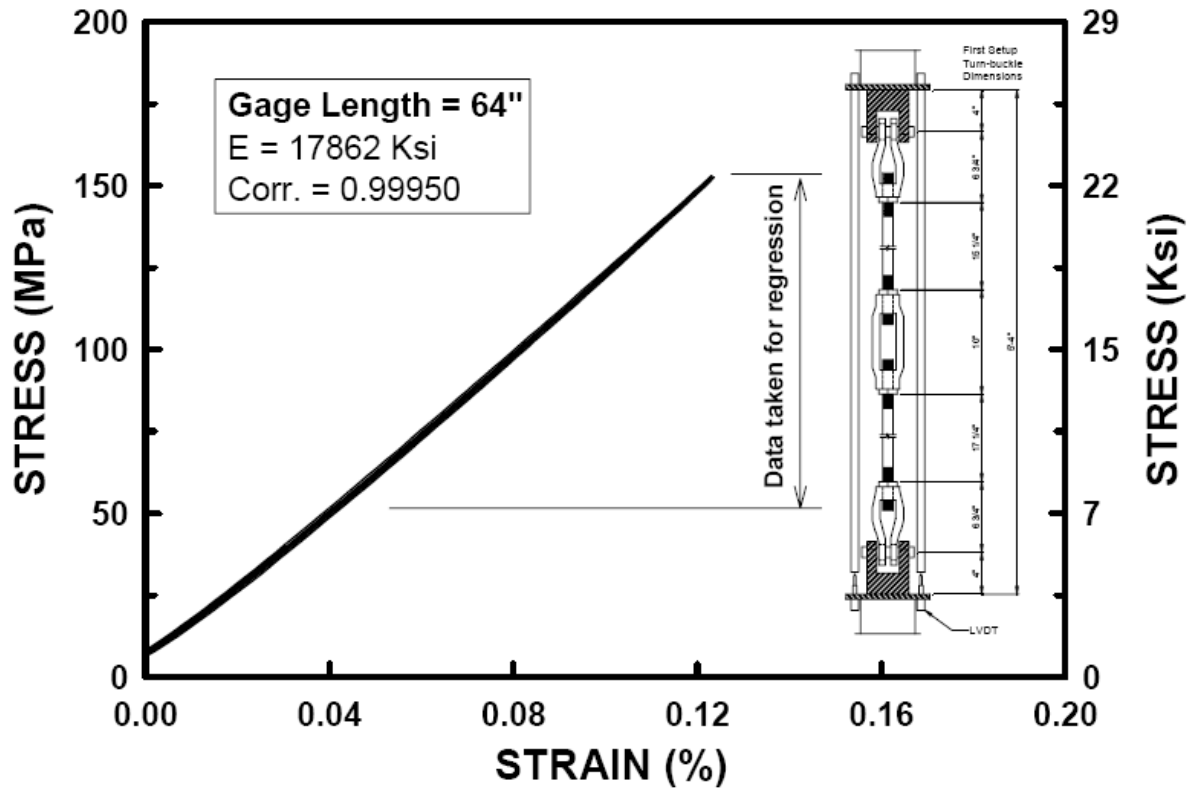


Figure A1 Stress-strain plot for hanger rod assemble test A.

